Running the primary experiment

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete research-case evidence record.

Report completion coverage, paired failures and later revisions without hiding missing cases.

Keep one habit: Recall → W1–W3/W5 → independent case → project evidence → feedback and delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Reproducible environment and records. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Pilot experiments. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Research ethics and consent. Give an example and a mistake that the idea helps you avoid.

Reproducible environment and records

> **My remembered idea, example and correction**

Pilot experiments

> **My remembered idea, example and correction**

Research ethics and consent

> **My remembered idea, example and correction**

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A primary experiment follows the fixed plan, preserves raw outcomes and reports deviations separately. Main twenty eligible cases include sixteen passes, two failures and two unassessed after outage. Completed count is eighteen. Planned completed-case pass rate is 16/18 ≈ 88.889%; coverage is 18/20 = 90%, with two unassessed outcomes explicit. The different 16/20 rate is not the same metric.

  Do not assume missing outcomes match completed ones or silently retry away failures. Separate frozen six-case picture has A flags 1, 0, 1, 1, 0, 1 and B flags 1, 1, 1, 0, 1, 1: totals four and five. B wins C2/C5, A wins C4, with three ties. Net gain one hides three changed cases. Tuning B using C4 creates a later revision; same-set rescoring is development evidence and the revised score is unknown.

  Supplied records establish no real model execution, significance or population advantage.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

- A primary experiment follows the fixed plan, preserves raw outcomes and reports deviations separately. Main twenty eligible cases include sixteen passes, two failures and two unassessed after outage. Completed count is eighteen. Planned completed-case pass rate is 16/18 ≈ 88.889%; coverage is 18/20 = 90%, with two unassessed outcomes explicit. The different 16/20 rate is not the same metric.

  Do not assume missing outcomes match completed ones or silently retry away failures. Separate frozen six-case picture has A flags 1, 0, 1, 1, 0, 1 and B flags 1, 1, 1, 0, 1, 1: totals four and five. B wins C2/C5, A wins C4, with three ties. Net gain one hides three changed cases. Tuning B using C4 creates a later revision; same-set rescoring is development evidence and the revised score is unknown.

  Supplied records establish no real model execution, significance or population advantage.

- Twenty eligible cases: 16 pass, 2 fail, 2 unassessed after outage. Plan reports completed rate plus coverage. Show the intermediate steps and give the conclusion with its scope.

Main quantity	Calculation	Interpretation
Eligible	16 + 2 + 2 = 20	All planned eligible cases
Completed	16 + 2 = 18	Assessed pass/fail outcomes
Completed-case pass rate	16/18 = 8/9 ≈ 88.889%	Rate among completed cases
Coverage	18/20 = 90%	Two of twenty remain unassessed
Unassessed	2	Retain as unknown outcomes
Different possible metric	16/20 = 80% known passes among all eligible	Not the planned completed-case rate

Main quantity	Calculation	Interpretation
Inference limit	Missingness may matter	Do not assume unassessed outcomes resemble completed cases

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A primary experiment follows the fixed plan, preserves raw outcomes and reports deviations separately. Main twenty eligible cases include sixteen passes, two failures and two unassessed after outage. Completed count is eighteen. Planned completed-case pass rate is 16/18 ≈ 88.889%; coverage is 18/20 = 90%, with two unassessed outcomes explicit. The different 16/20 rate is not the same metric.

  Do not assume missing outcomes match completed ones or silently retry away failures. Separate frozen six-case picture has A flags 1, 0, 1, 1, 0, 1 and B flags 1, 1, 1, 0, 1, 1: totals four and five. B wins C2/C5, A wins C4, with three ties. Net gain one hides three changed cases. Tuning B using C4 creates a later revision; same-set rescoring is development evidence and the revised score is unknown.

  Supplied records establish no real model execution, significance or population advantage.

- A report omits this qualification: Following the plan and reporting deviations are required even when results contradict the hypothesis.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A primary experiment follows the fixed plan, preserves raw outcomes and reports deviations separately. Main twenty eligible cases include sixteen passes, two failures and two unassessed after outage. Completed count is eighteen. Planned completed-case pass rate is 16/18 ≈ 88.889%; coverage is 18/20 = 90%, with two unassessed outcomes explicit. The different 16/20 rate is not the same metric.

  Do not assume missing outcomes match completed ones or silently retry away failures. Separate frozen six-case picture has A flags 1, 0, 1, 1, 0, 1 and B flags 1, 1, 1, 0, 1, 1: totals four and five. B wins C2/C5, A wins C4, with three ties. Net gain one hides three changed cases. Tuning B using C4 creates a later revision; same-set rescoring is development evidence and the revised score is unknown.

  Supplied records establish no real model execution, significance or population advantage.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Twelve eligible cases include 9 passes, 1 failure and 2 unassessed cases. Report completed-case pass rate (passes / completed cases), assessment coverage (completed / eligible cases) and the unassessed count.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled running the primary experiment **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of running the primary experiment.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Running the primary experiment. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Use the worked case for Running the primary experiment. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Twelve eligible cases include 9 passes, 1 failure and 2 unassessed cases. Report completed-case pass rate (passes / completed cases), assessment coverage (completed / eligible cases) and the unassessed count.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

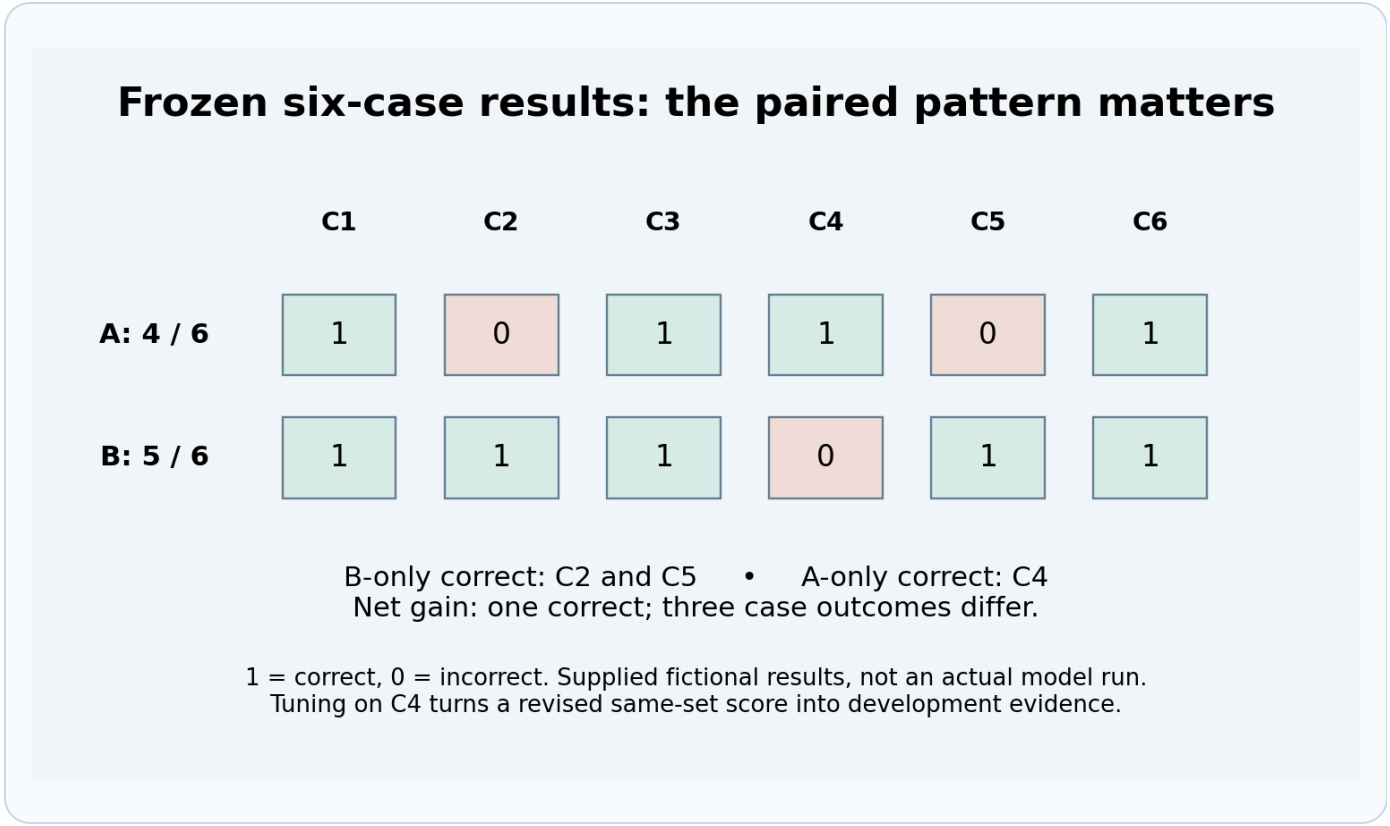
- Run the frozen experiment. A research team freezes versions A and B, labels and scoring before a six-case **primary run**.

- A scores [1,0,1,1,0,1]; B scores [1,1,1,0,1,1]. Both answer the same six cases. Binary entries mean 1=correct and 0=incorrect.

- Compute A's and B's six-case counts and identify the cases that favour each. How must a B revision tuned on case four be labelled when evaluated again on the same set?

- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.



AI ACADEMY / AI-7/25

Paired primary case outcomes

	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6
Version A	1	0	1	1	0	1
Version B	1	1	1	0	1	1

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/25

Running the primary experiment

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Keep versions fixed

No case result may trigger tuning during this run.



2. Compute results

A totals 4/6; B totals 5/6.



3. Keep case-level records

Cases two and five favour B; case four favours A.

CHANGE ONE THING

Tune B after inspecting its case-four failure, then rescore the same set.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Run the frozen experiment. A research team freezes versions A and B, labels and scoring before a six-case **primary run**.
- Which labelled step matches this detail? Cases two and five favour B; case four favours A.
- 1. Compute results
- 2. Keep versions fixed
- 3. Keep case-level records
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Keep case-level records Cases two and five favour B; case four favours A.

Running the primary experiment means following the declared protocol and retaining raw outcomes. Overall scores alone hide different failure patterns.

Changed-case question and optional hint

- Changed-case practice: Tune B after inspecting its case-four failure, then rescore the same set. Which conclusion is supported by the supplied facts?
- 1. Label the rerun as development on exposed cases, not untouched primary evaluation.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Label the rerun as development on exposed cases, not untouched primary evaluation. The evaluation outcomes have now influenced the version.

Repair and retain the record

Review Reproducible environment and records (ai-7/24). Goal: Explain the **mechanism** and terminology of reproducible environment and records. Short practice: Trace the recorded mean for 2,4,6 and its changed result for 2,4,9.

Which artefacts must accompany the **output** so another team can identify the exact run?. Return to this week and explain what you changed.

Review Pilot experiments (ai-7/22). Goal: Explain the **mechanism** and terminology of pilot experiments. Short practice: Calculate reviewer agreement on the four pilot answers.

What can the disagreement improve, and why are those revised practice answers not untouched final **evidence**?. Return to this week and explain what you changed.

Review Research ethics and consent (ai-7/19). Goal: Explain the **mechanism** and terminology of research ethics and consent. Short practice: Identify which uses the drawing consent permits and excludes.

Can the unrelated project reuse the drawings under this agreement, and what new authorisation would be needed?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Running the primary experiment: the case

AI ACADEMY / AI-7/25 · 01 · LOOK AT THE FACTS

Running the primary experiment: the case

The situation

Run the frozen experiment. A research team freezes versions A and B, labels and scoring before a six-case primary run.

Supplied evidence

A scores [1,0,1,1,0,1]; B scores [1,1,1,0,1,1]. Both answer the same six cases. Binary entries mean 1=correct and 0=incorrect.

Explain running the primary experiment using the records and boundaries of this fictional investigation.

Follow the labelled picture with the matching explanation. Open the vector diagram

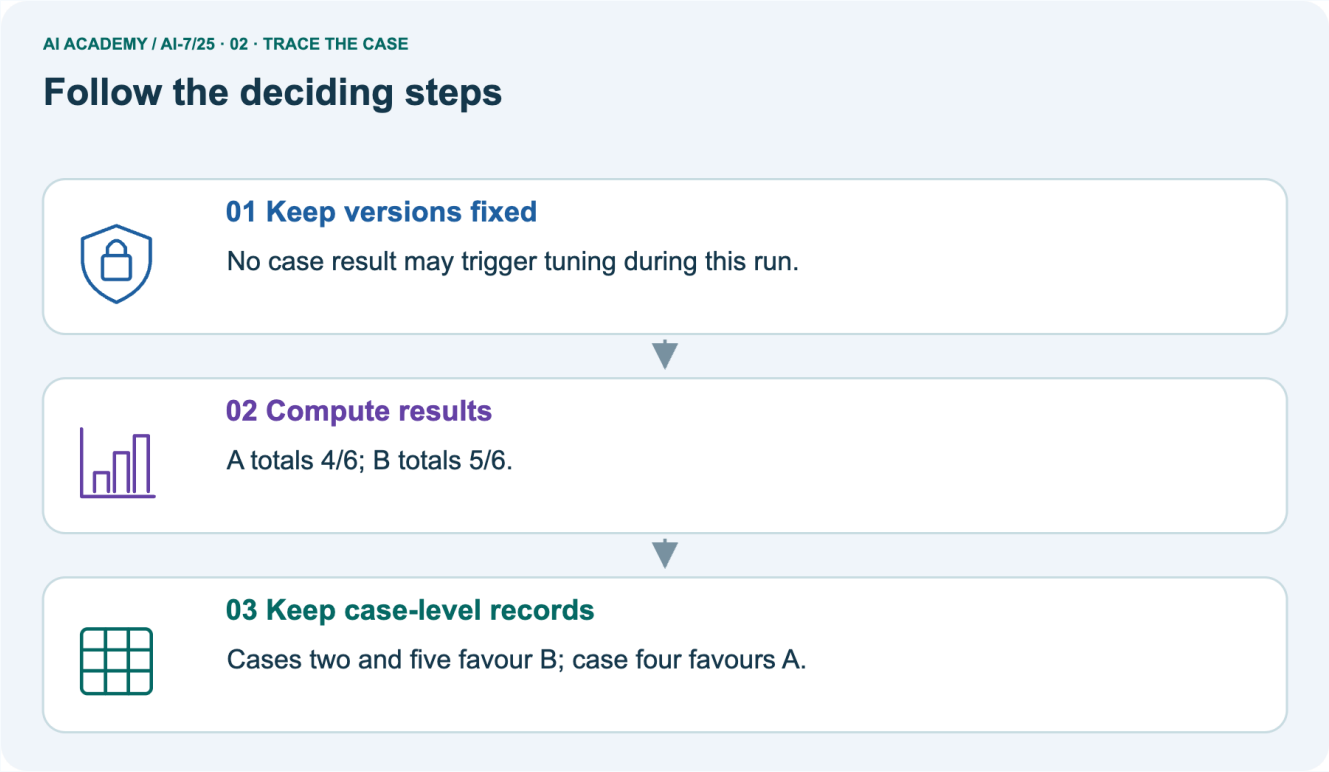
Run the frozen experiment. A research team freezes versions A and B, labels and scoring before a six-case **primary run**. Supplied facts: A scores [1,0,1,1,0,1]; B scores [1,1,1,0,1,1]. Both answer the same six cases.

Binary entries mean 1=correct and 0=incorrect.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



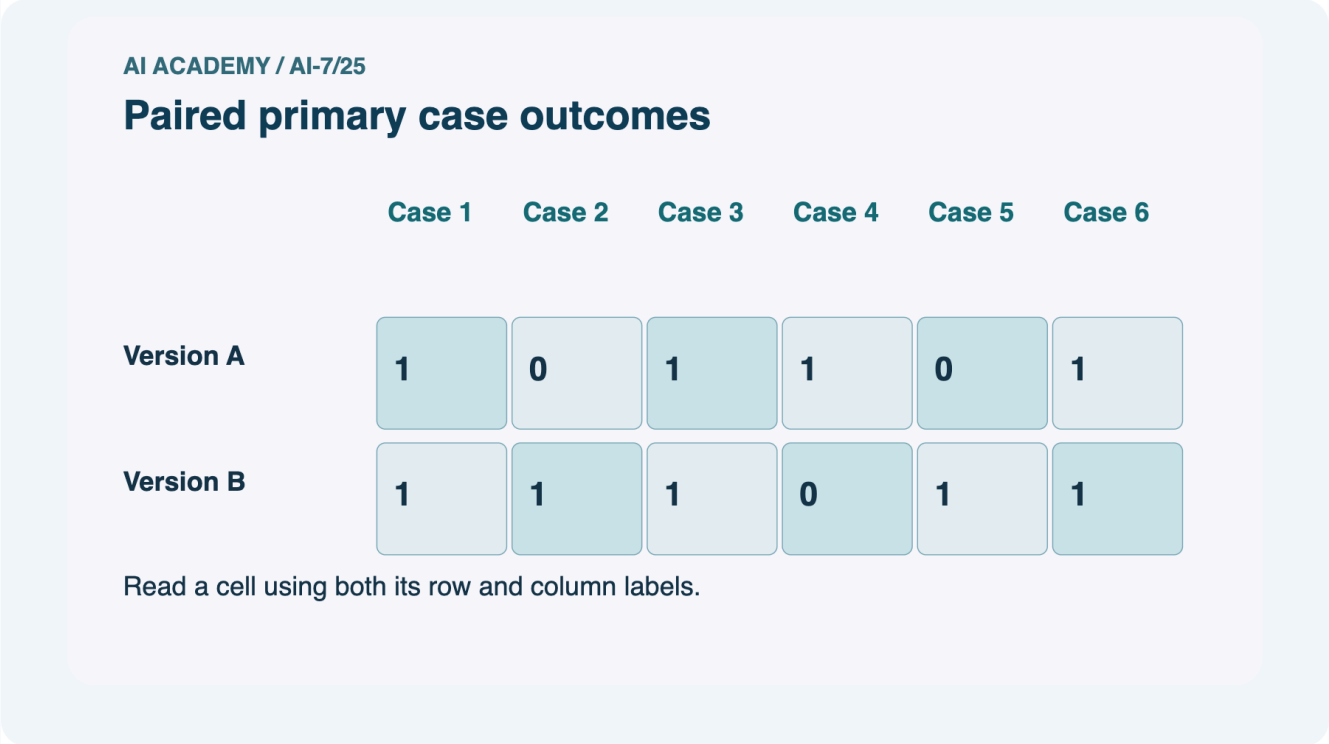
Follow the labelled picture with the matching explanation. Open the vector diagram

Keep versions fixed: No case result may trigger tuning during this run. → Compute results: A totals 4/6; B totals 5/6. → Keep case-level records: Cases two and five favour B; case four favours A.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Paired primary case outcomes



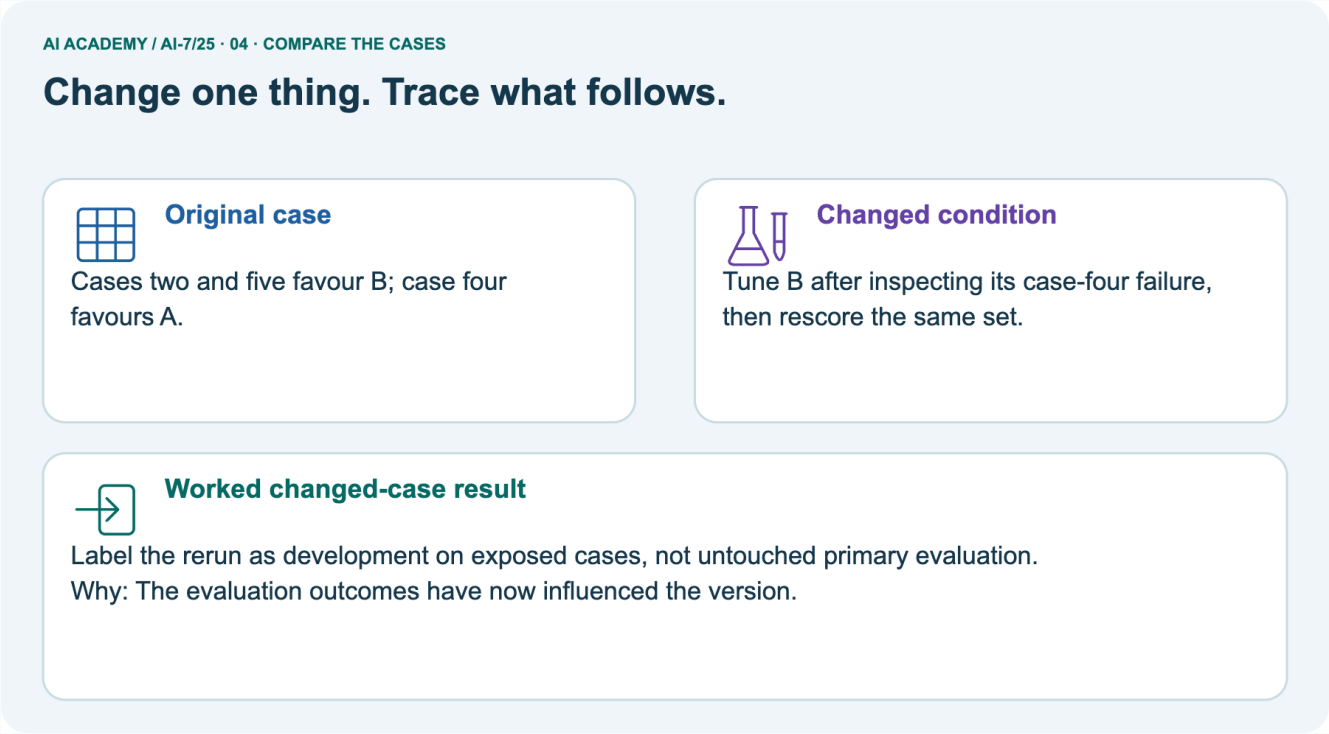
Follow the labelled picture with the matching explanation. Open the vector diagram

Paired primary case outcomes. Version A: Case 1 = 1, Case 2 = 0, Case 3 = 1, Case 4 = 1, Case 5 = 0, Case 6 = 1. Version B: Case 1 = 1, Case 2 = 1, Case 3 = 1, Case 4 = 0, Case 5 = 1, Case 6 = 1.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

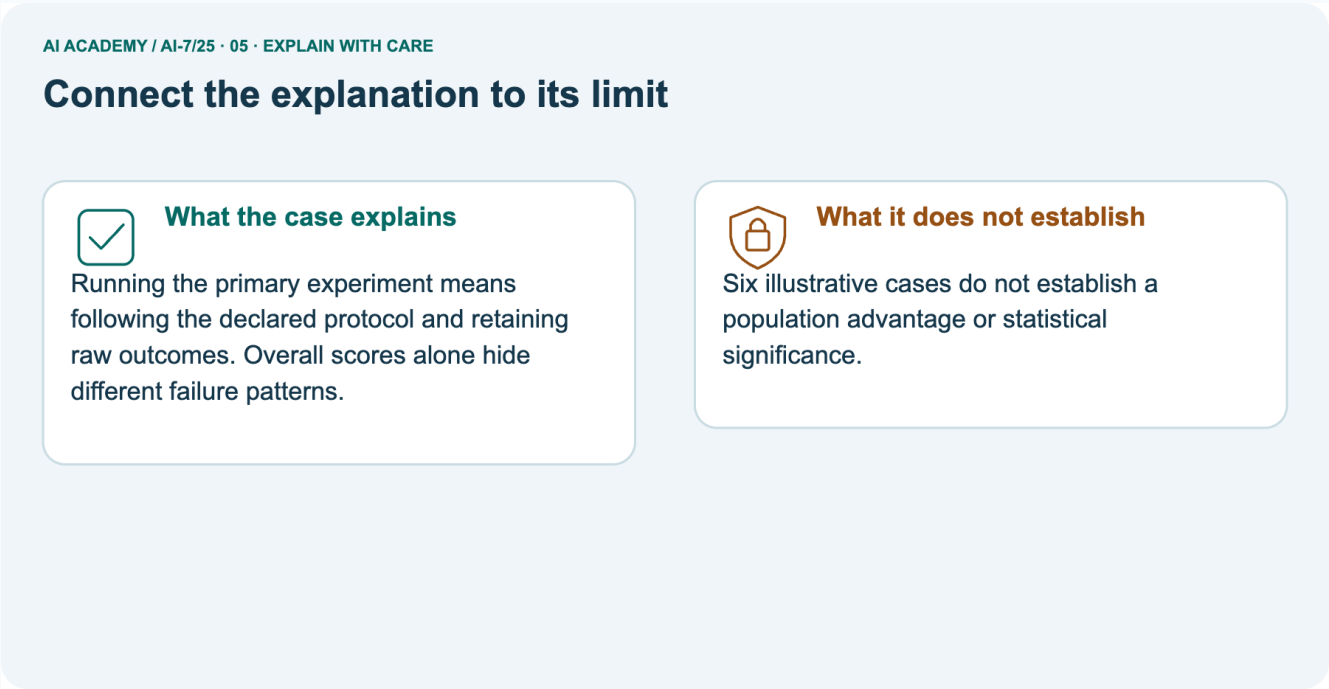
Original case: Cases two and five favour B; case four favours A. Changed condition: Tune B after inspecting its case-four failure, then rescore the same set.

Expected result: Label the rerun as development on exposed cases, not untouched primary evaluation. Reason: The evaluation outcomes have now influenced the version.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

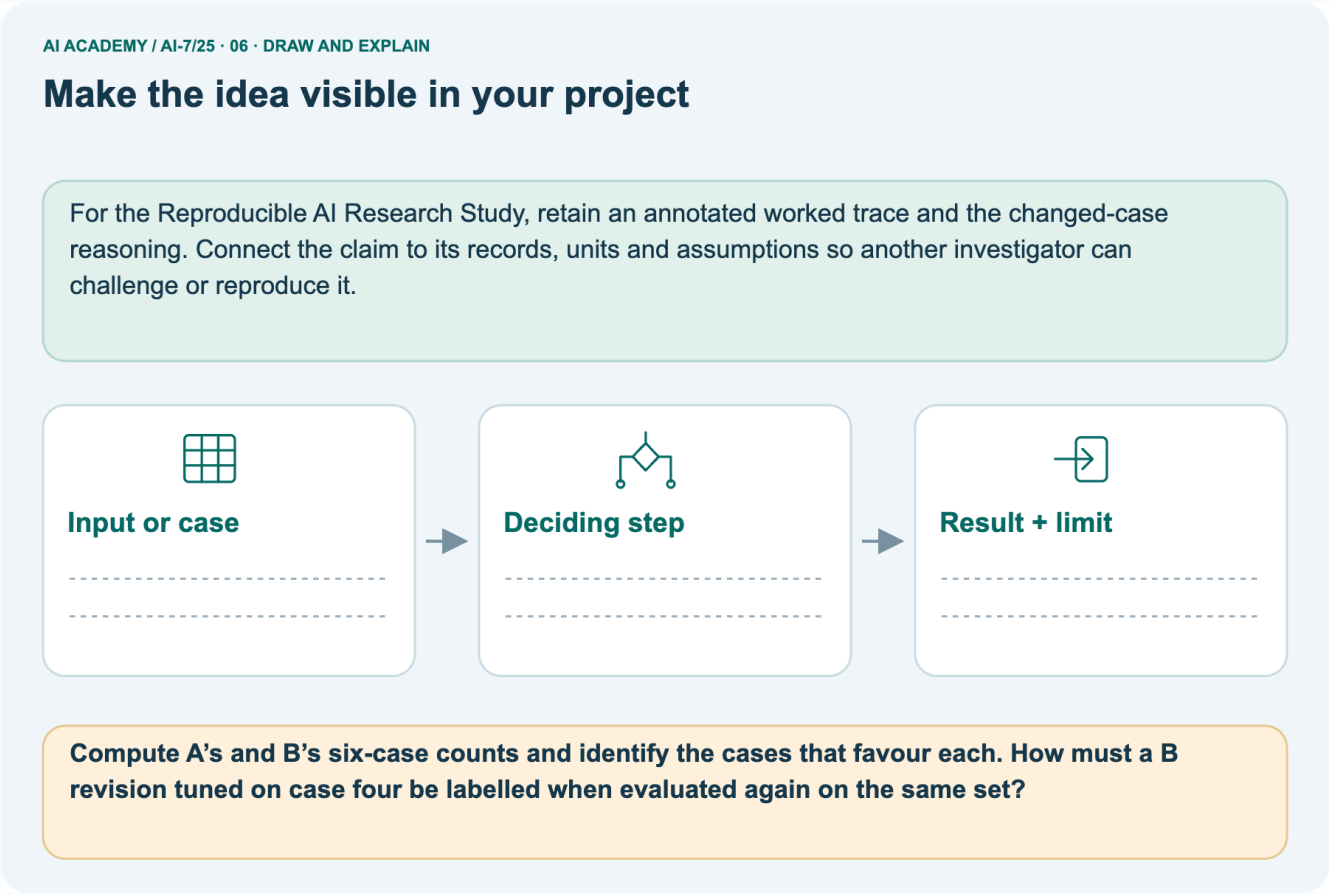
Running the primary experiment means following the declared protocol and retaining raw outcomes. Overall scores alone hide different failure patterns.

Limit: Six illustrative cases do not establish a population advantage or statistical significance.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Compute A's and B's six-case counts and identify the cases that favour each.

How must a B revision tuned on case four be labelled when evaluated again on the same set?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Running the primary experiment: the case

02. Follow the deciding steps

03. Paired primary case outcomes

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.